Yi Yang

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Summary

CMU Ph.D. candidate applying machine learning with first-principles (DFT) simulation for molecular crystal structure prediction. Train, fine-tune, and evaluate graph neural network potentials and build flow matching models in PyTorch. With Meta FAIR Chemistry, co-created the 27M+ structure OMC25 dataset that trained the foundation model and co-developed the open-source FastCSP pipeline to accelerate materials discovery. Lead developer of the Genarris structure generator (Python/C). Graduating December 2026; available January or February 2027.

Education

Carnegie Mellon University — Pittsburgh, PA
Doctor of Philosophy (Ph.D.) in Materials Science and Engineering — *Jan. 2023 – Dec. 2026 (expected)*
- Research: AI for molecular crystal structure prediction. Advisor: Noa Marom.

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Computational Materials Science and Engineering — *Aug. 2021 – Dec. 2022*

China University of Geosciences, Wuhan — Wuhan, China
Bachelor of Science in Materials Science and Engineering — *Sept. 2017 – July 2021*

Research Experience

Graduate Researcher, CMU–Meta FAIR Chemistry Collaboration — Pittsburgh, PA
Industry Research Collaboration: Foundation Model Training Data, Fine-Tuning, and Inference — *2024 – Present*

OMC25 and UMA: Training Data, Model Training, and Evaluation
- Co-created **Open Molecular Crystals 2025 (OMC25)**, a **27M+**-structure dataset used to train Meta's **Universal Model for Atoms (UMA)**, a **graph neural network** foundation model. Labels include dispersion-inclusive density functional theory (DFT) properties energies, forces, and stresses.
- Contributed to UMA **post-training** for molecular crystal tasks, and evaluated each round of trained models with FAIR collaborators on the held-out test split and external benchmarks.

FastCSP: Accelerated Crystal Structure Prediction with a Foundation Model | Source
- Co-developed **FastCSP** in Meta's open-source fairchem Python library to support material discovery: an end-to-end workflow that predicts a molecule's crystal structures on one pretrained UMA model, from conformer and structure generation (Genarris) through relaxation and free-energy ranking.
- Accelerated **scientific discovery** by enabling crystal structure prediction within hours on GPU, replacing classical force-field screening and DFT re-ranking with pretrained UMA model without system-specific fine-tuning.
- Benchmarked FastCSP and recovered all **experimental known crystal structures** and ranked each within **9 kJ/mol** of the global minimum.

Graduate Researcher, Carnegie Mellon University — Pittsburgh, PA
Machine Learning for Crystal Structure Prediction; Advisor: Noa Marom — *Sept. 2021 – Present*

Generative Modeling with Flow Matching
- Developing a space-group-conditional **flow matching** model in PyTorch for de novo crystal structure prediction.
- Developing adjoint sampling fine-tuning, a stochastic optimal control method, against multi-fidelity rewards from ML-potential relaxation, to target greater stability and physical validity of generated structures.

Genarris: Open-Source Structure Generation and Optimization | GitHub
- Lead developer of **Genarris 3.0/4.0**, a Python/C package with MPI parallelism that samples random crystal structures across space groups and generates **millions of candidates** on HPC clusters. The code produced the structures behind OMC25 and FastCSP.
- Developed a **geometry optimization algorithm** that close-packs unit cells with a regularized hard-sphere potential, with **up to 100**× lower computational cost than ML-potential relaxation in published benchmarks.
- Reduced the candidate pool to roughly **10%** using **unsupervised clustering** while retaining experimental matches.

Seventh Blind Test of Crystal Structure Prediction (Cambridge Crystallographic Data Center)
- Contributed predictions in this international blind challenge to our team's **67% success rate**, **first among academic teams and third overall**.
- Developed system-specific AIMNet2 neural network potentials as DFT surrogates using active learning, with custom data sampling and augmentation; the models matched DFT accuracy at a fraction of the computational cost.

Selected Publications

[1] **Yang, Y.**, et al. Structure Prediction of Multi-Component Molecular Crystals with Genarris 4.0. *ChemRxiv* (2026).

[2] Gharakhanyan, V.*, **Yang, Y.***, et al. FastCSP: Accelerated Molecular Crystal Structure Prediction with Universal Model for Atoms. *arXiv:2508.02641* (2025). [∗ Equal contribution.]

[3] Gharakhanyan, V., **Yang, Y.**, et al. Open Molecular Crystals 2025 (OMC25) Dataset and Models. *Scientific Data*, **13**, 354 (2026).

[4] **Yang, Y.**, et al. Genarris 3.0: Generating Close-Packed Molecular Crystal Structures with Rigid Press. *J. Chem. Theory Comput.* **21**, 11318–11332 (2025).

[5] Nayal, K. S., **Yang, Y.**, et al. Efficient Molecular Crystal Structure Prediction and Stability Assessment with AIMNet2 Neural Network Potentials. *Crystal Growth & Design* **25**, 9092–9106 (2025).

[6] Hunnisett, L. M., **Yang, Y.**, et al. The Seventh Blind Test of Crystal Structure Prediction: Structure Ranking Methods. *Acta Cryst. B* **80**, 548–574 (2024).

[7] Tom, R., **Yang, Y.**, et al. Inverse Design of Tetracene Polymorphs with Enhanced Singlet Fission Performance by Property-Based Genetic Algorithm Optimization. *Chemistry of Materials* **35**, 1373–1386 (2023).

Selected Projects

CardAgent: LLM Planning with Retrieval and Verification | *11-766, CMU* Jan. 2026 – May 2026

- Built an LLM planner with a deterministic Python verifier and expected-value scoring for constrained sequential decisions; a **6-month** simulator searched action sequences under **8 constraints**.
- Built retrieval-augmented generation (RAG) with ChromaDB and sentence embeddings that drops expired evidence before retrieval. The deterministic checks caught rule-violating plans that baseline LLM planners let through.

Technical Skills

Programming: Python, C, MPI
Machine learning: PyTorch, graph neural network, generative models (flow matching, diffusion), Transformers, active learning
LLM applications: RAG, ChromaDB, sentence embeddings, LLM planning with deterministic verification
First-principles simulation: dispersion-inclusive DFT (FHI-aims, VASP, Orca)
Computing: GPU inference, Slurm, HPC clusters (Bridges2, Frontera, Vista, Perlmutter)

Honors

Presidential Fellowship, College of Engineering, CMU	*Feb. 2026*
Machine Learning in Chemical and Materials Sciences (MLCM) Rising Star Award, LANL	*May 2025*
Frontera Computational Science Fellowship, TACC	*Apr. 2025*
Award for Research Excellence in the Master Program, CMU	*May 2023*
ATK-Nick G. Vlahakis Graduate Fellowship, CMU	*Feb. 2023*
Outstanding Undergraduate Award, CUG	*June 2021*